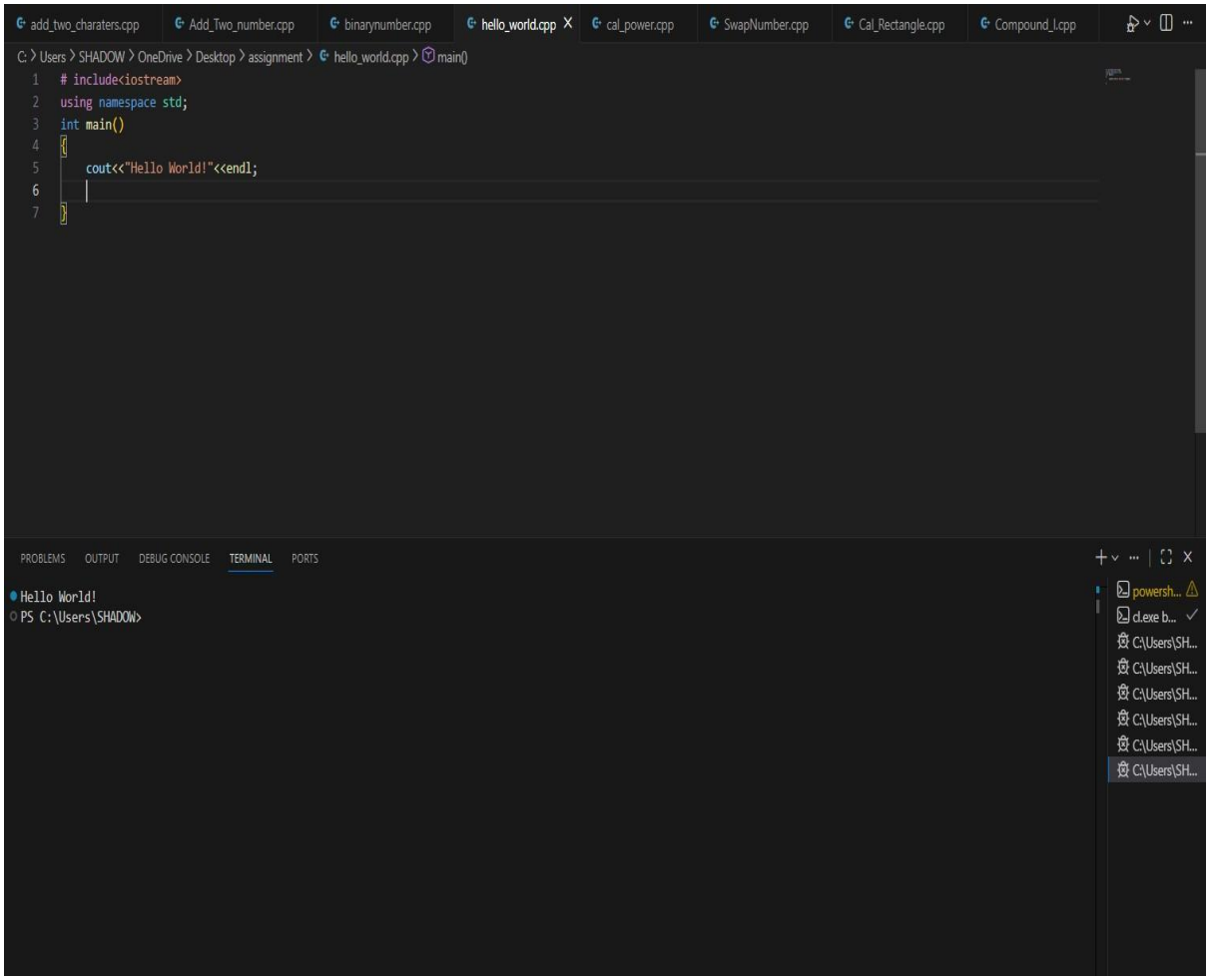
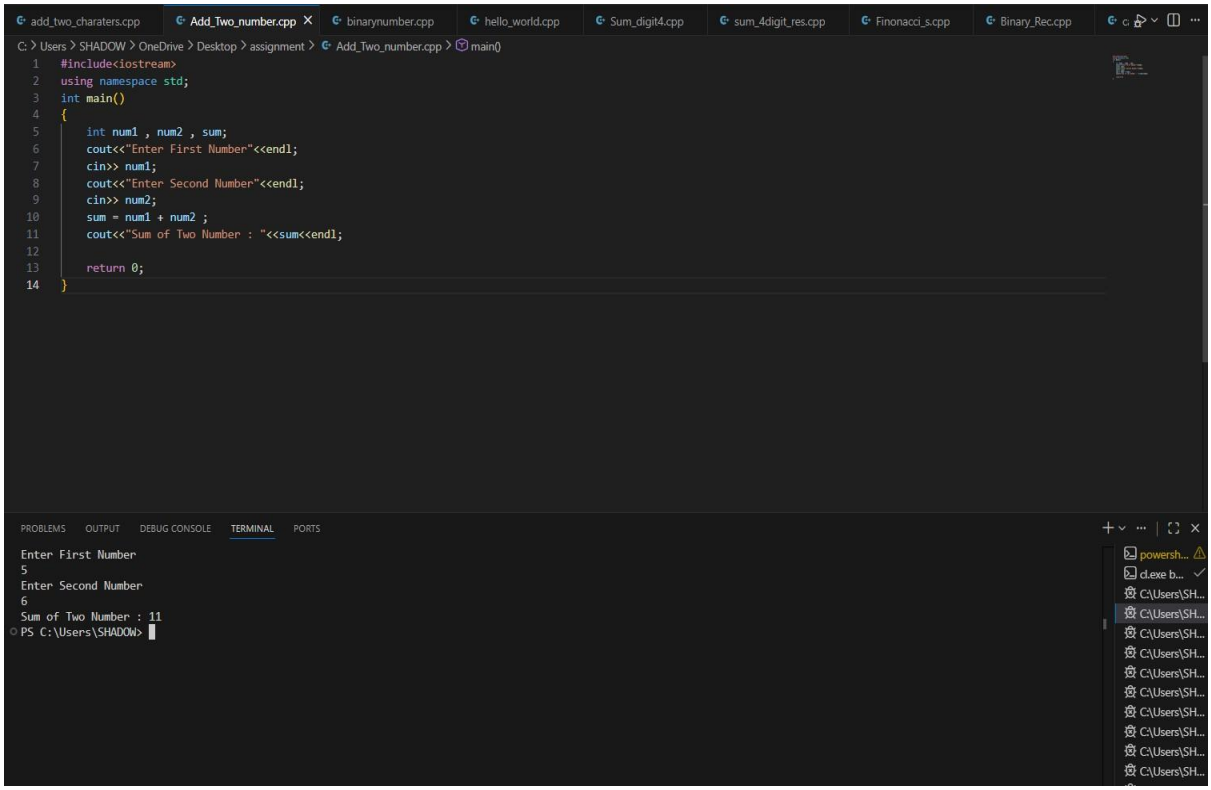
Q1]Print Hello World

```
# include<iostream>
using namespace std;
int main()
{
   cout<<"Hello World!"<<endl;
}
```



Q2] Add two Number /Binary number/Characters.

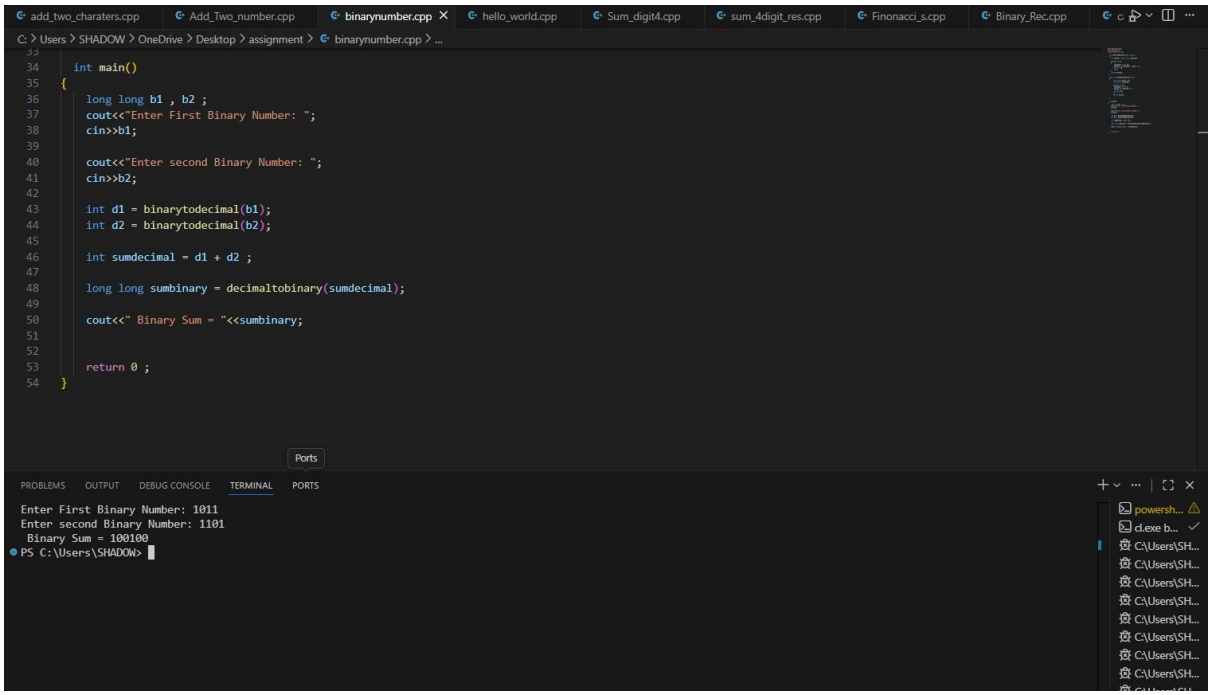
```
#include<iostream>
using namespace std;
int main()
{
   int num1 , num2 , sum;
   cout<<"Enter First Number"<<endl;
   cin>> num1;
   cout<<"Enter Second Number"<<endl;
   cin>> num2;
   sum = num1 + num2 ;
   cout<<"Sum of Two Number : "<<sum<<endl;

   return 0;
}
```



Q2] Binary number

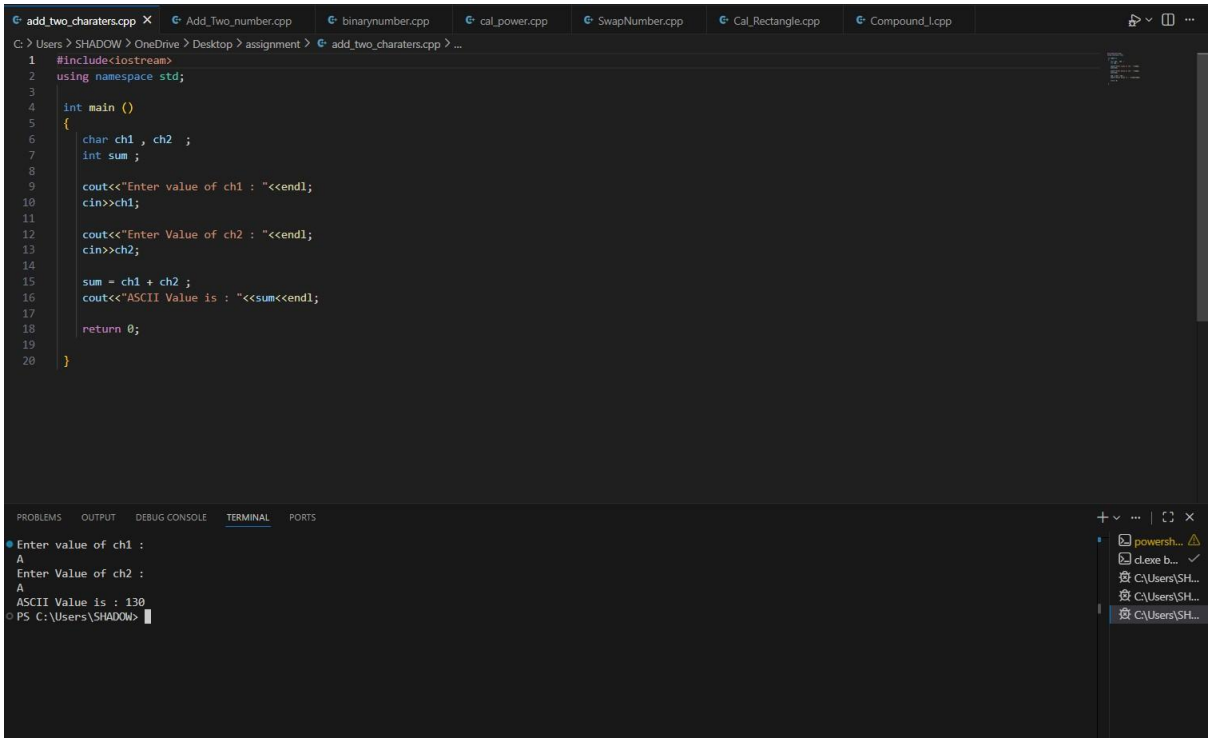
```
#include<iostream>
#include<math.h>
using namespace std;

 int binarytodecimal(long long n)
 {
   int decimal = 0, i = 0, remainder;

   while(n != 0)
   {
      remainder = n % 10;
      decimal += remainder + pow(2, i);
      n = n / 10;
      i++;
   }
   return decimal;
 }

 long long decimaltobinary(int n)
 {
     long long binary = 0;
     int i = 1, remainder;

     while(n != 0){
      remainder = n % 2;
      binary += remainder * i ;
      n = n / 2 ;
      i = i * 10 ;
     }
     return binary;
```

```
 }

 int main()
{
   long long b1 , b2 ;
   cout<<"Enter First Binary Number: ";
   cin>>b1;

   cout<<"Enter second Binary Number: ";
   cin>>b2;

   int d1 = binarytodecimal(b1);
   int d2 = binarytodecimal(b2);
   int sumdecimal = d1 + d2 ;
   long long sumbinary = decimaltobinary(sumdecimal);
   cout<<" Binary Sum = "<<sumbinary;
   return 0 ;
}
```



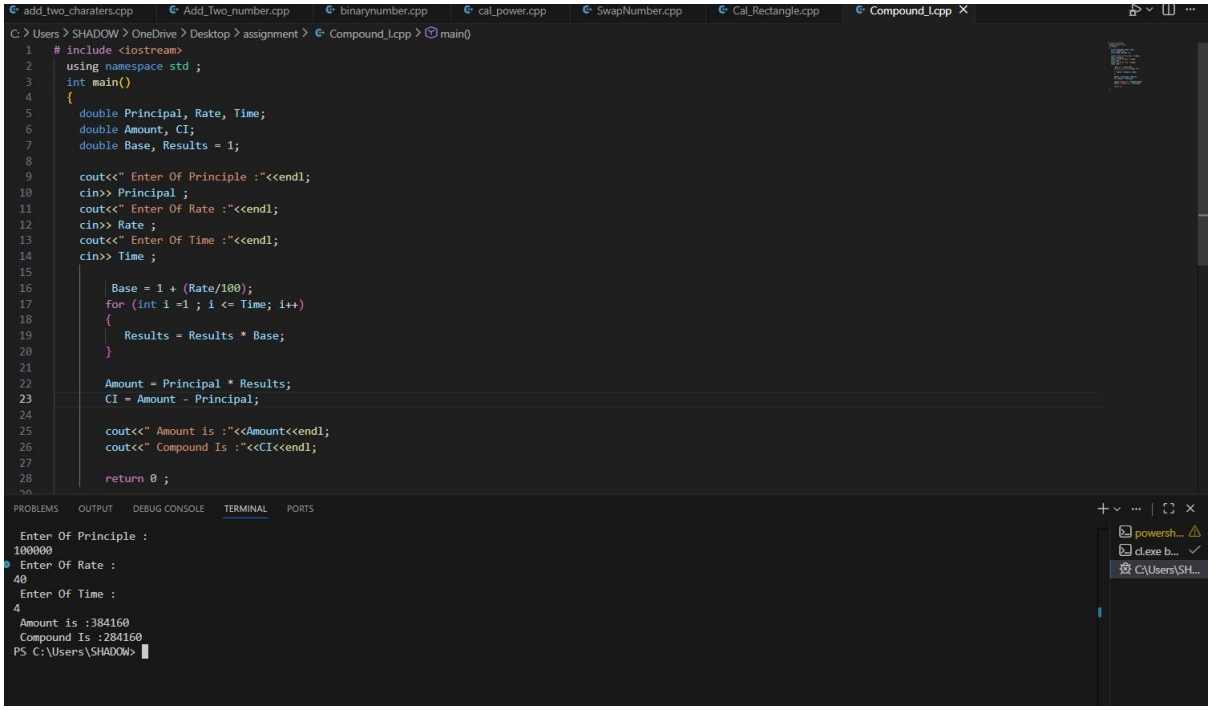
Q2] Add Two Character

```
#include<iostream>
using namespace std;

 int main ()
{
   char ch1 , ch2  ;
   int sum ;

   cout<<"Enter value of ch1 : "<<endl;
   cin>>ch1;
   cout<<"Enter Value of ch2 : "<<endl;
   cin>>ch2;
   sum = ch1 + ch2 ;
   cout<<"ASCII Value is : "<<sum<<endl;
   return 0;
}
```



Q3] Cal Compound Interest

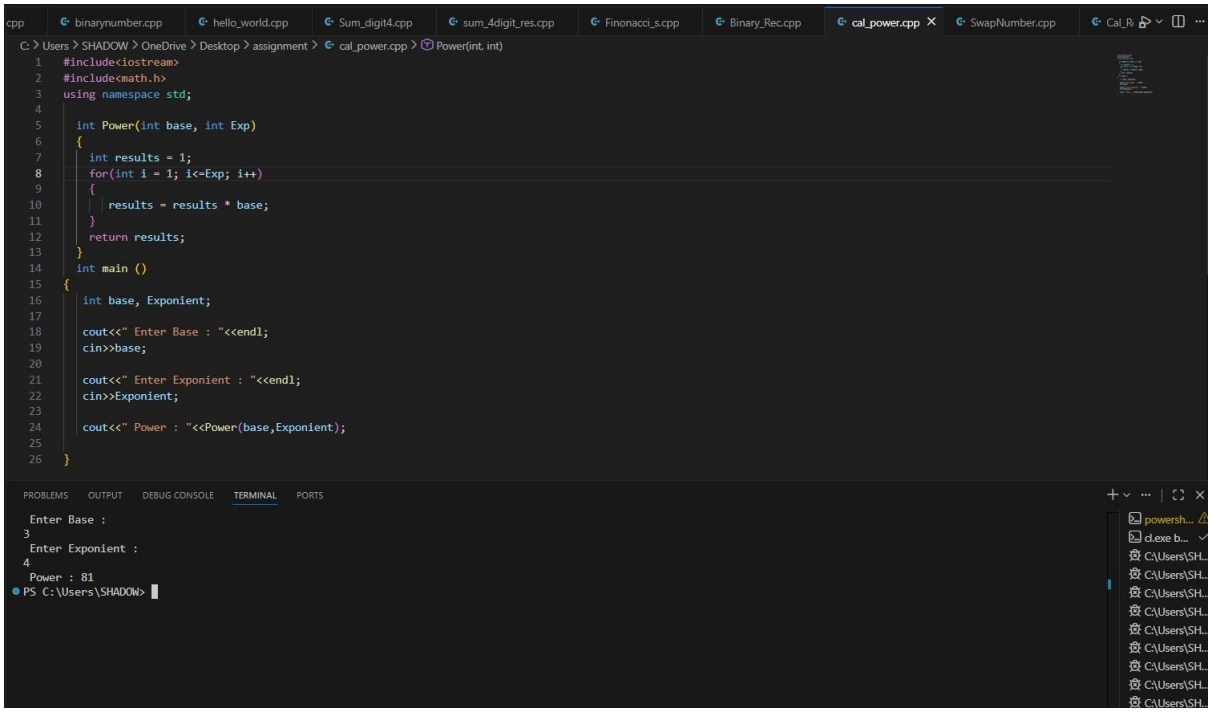
```
# include <iostream>
 using namespace std ;
 int main()
 {
   double Principal, Rate, Time;
   double Amount, CI;
   double Base, Results = 1;

   cout<<" Enter Of Principle :"<<endl;
   cin>> Principal ;
   cout<<" Enter Of Rate :"<<endl;
   cin>> Rate ;
   cout<<" Enter Of Time :"<<endl;
   cin>> Time ;

      Base = 1 + (Rate/100);
     for (int i =1 ; i <= Time; i++)
     {
       Results = Results * Base;
     }

     Amount = Principal * Results;
     CI = Amount - Principal;

     cout<<" Amount is :"<<Amount<<endl;
     cout<<" Compound Is :"<<CI<<endl;

     return 0 ;

 }
```



Q4] cal Power

```
#include<iostream>
#include<math.h>
using namespace std;

 int Power(int base, int Exp)
 {
  int results = 1;
  for(int i = 1; i<=Exp; i++)
  {
    results = results * base;
  }
  return results;
 }
 int main ()
{
  int base, Exponient;
```

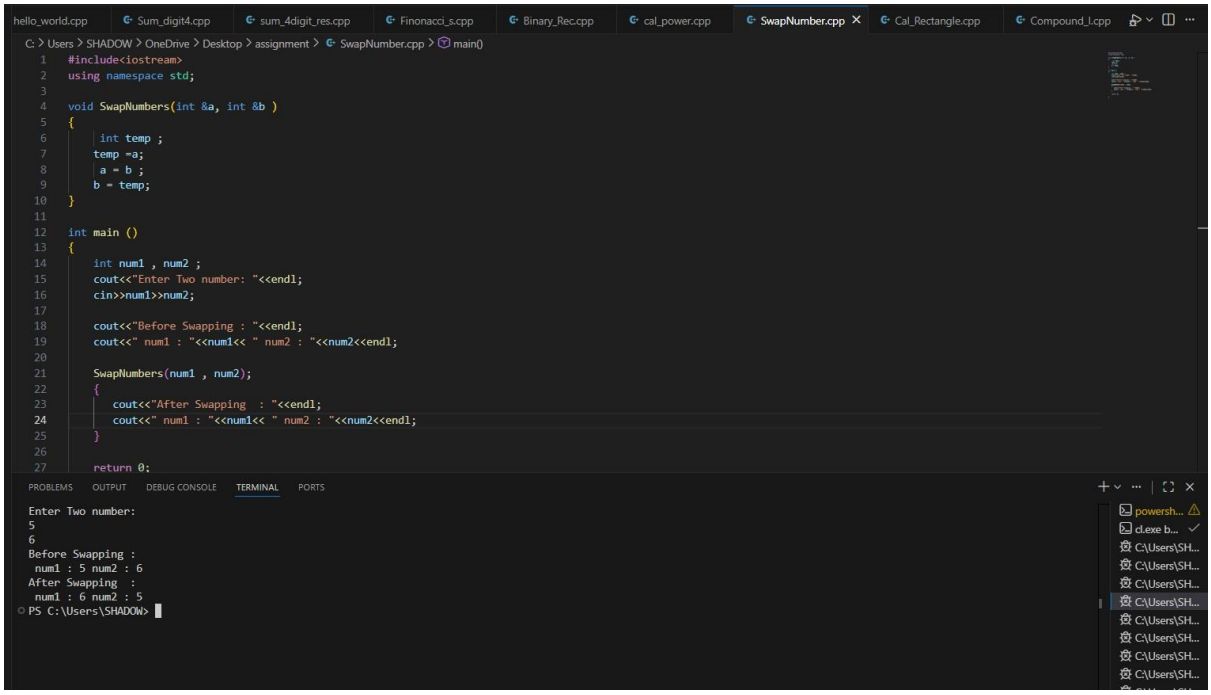
```
  cout<<" Enter Base : "<<endl;
  cin>>base;

  cout<<" Enter Exponient : "<<endl;
  cin>>Exponient;

  cout<<" Power : "<<Power(base,Exponient);

}
```



Q5] Swapping two number

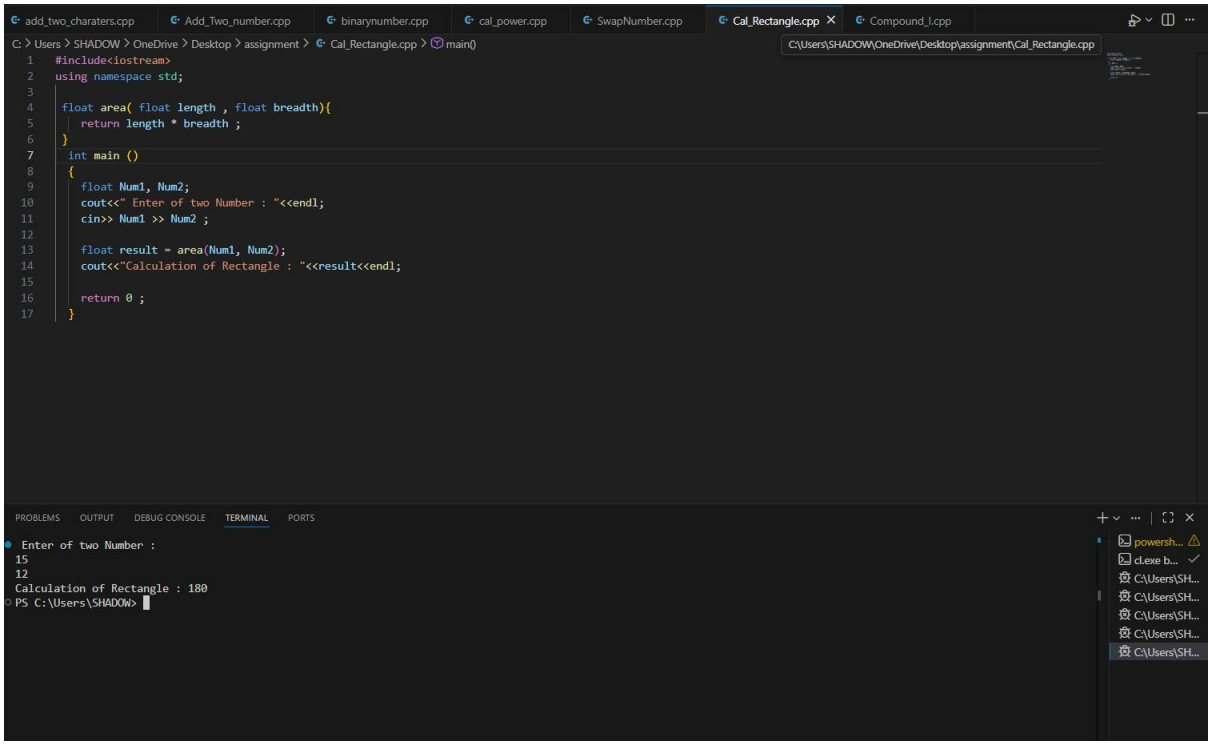
```
#include<iostream>
using namespace std;

void SwapNumbers(int &a, int &b )
{
    int temp ;
   temp =a;
```

```
    a = b ;
   b = temp;
}

int main ()
{
   int num1 , num2 ;
   cout<<"Enter Two number: "<<endl;
   cin>>num1>>num2;

   cout<<"Before Swapping : "<<endl;
   cout<<" num1 : "<<num1<< " num2 : "<<num2<<endl;

   SwapNumbers(num1 , num2);
   {
     cout<<"After Swapping  : "<<endl;
     cout<<" num1 : "<<num1<< " num2 : "<<num2<<endl;
   }

   return 0;

}
```



Q6] cal Area of Rectangle

```
#include<iostream>
using namespace std;

 float area( float length , float breadth){
   return length * breadth ;
 }
  int main ()
  {
   float Num1, Num2;
   cout<<" Enter of two Number : "<<endl;
   cin>> Num1 >> Num2 ;

   float result = area(Num1, Num2);
   cout<<"Calculation of Rectangle : "<<result<<endl;

   return 0 ;
  }
```

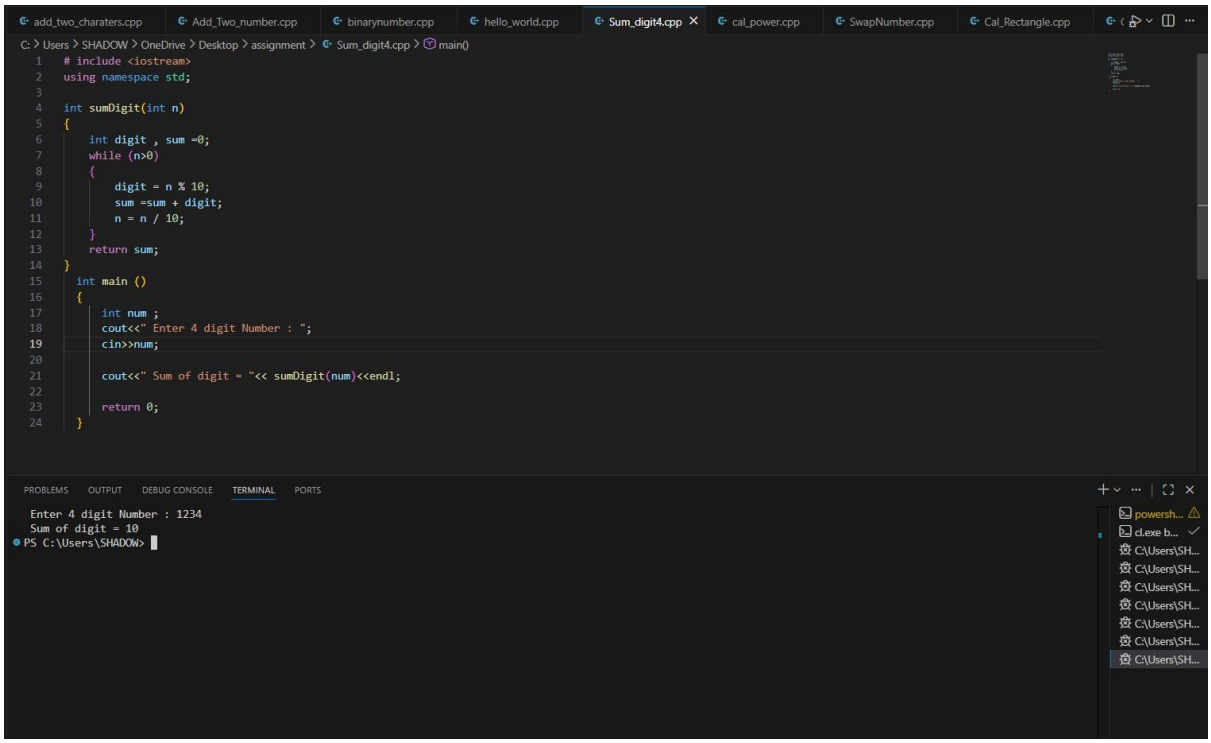


Q7] 4 digit number using

Loop

```
# include <iostream>

using namespace std;


int sumDigit(int n)

{

   int digit , sum =0;

   while (n>0)

   {

      digit = n % 10;

      sum =sum + digit;

      n = n / 10;

   }

   return sum;

}

 int main ()
```

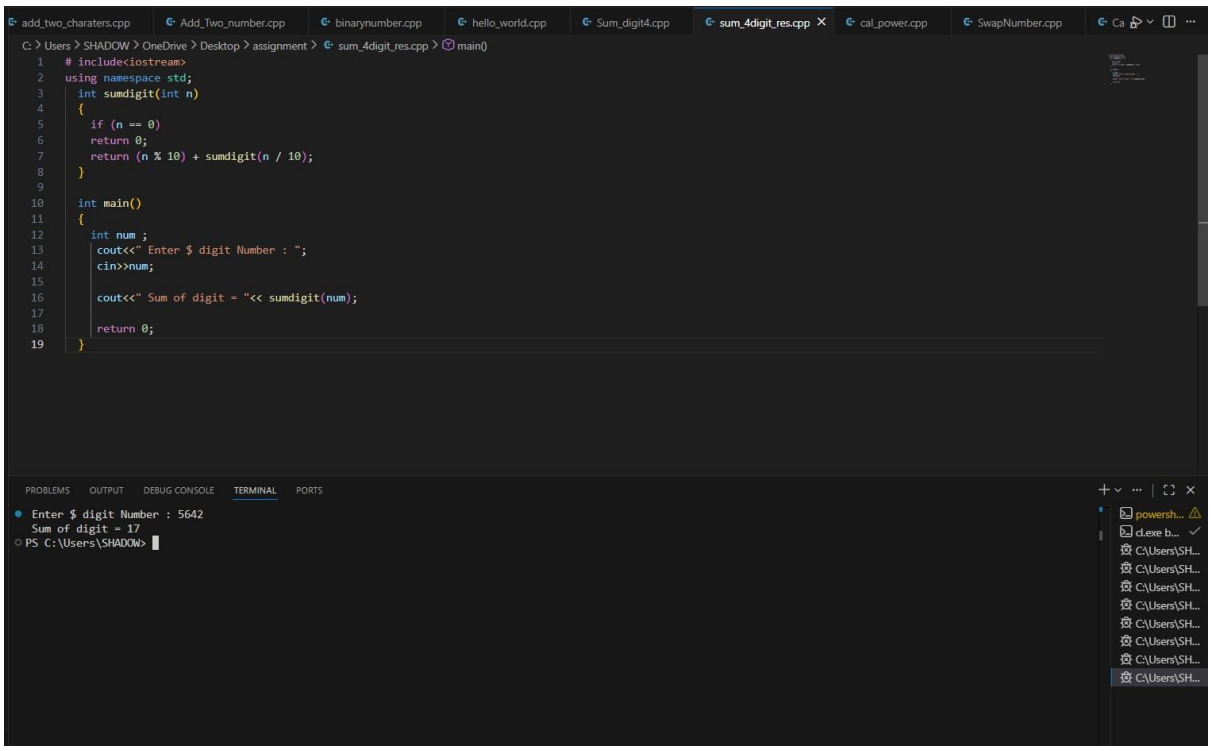
```
{
    int num ;
    cout<<" Enter 4 digit Number : ";
    cin>>num;

    cout<<" Sum of digit = "<< sumDigit(num)<<endl;

    return 0;
}
```



Recursion

```
# include<iostream>
using namespace std;
 int sumdigit(int n)
 {
  if (n == 0)
```

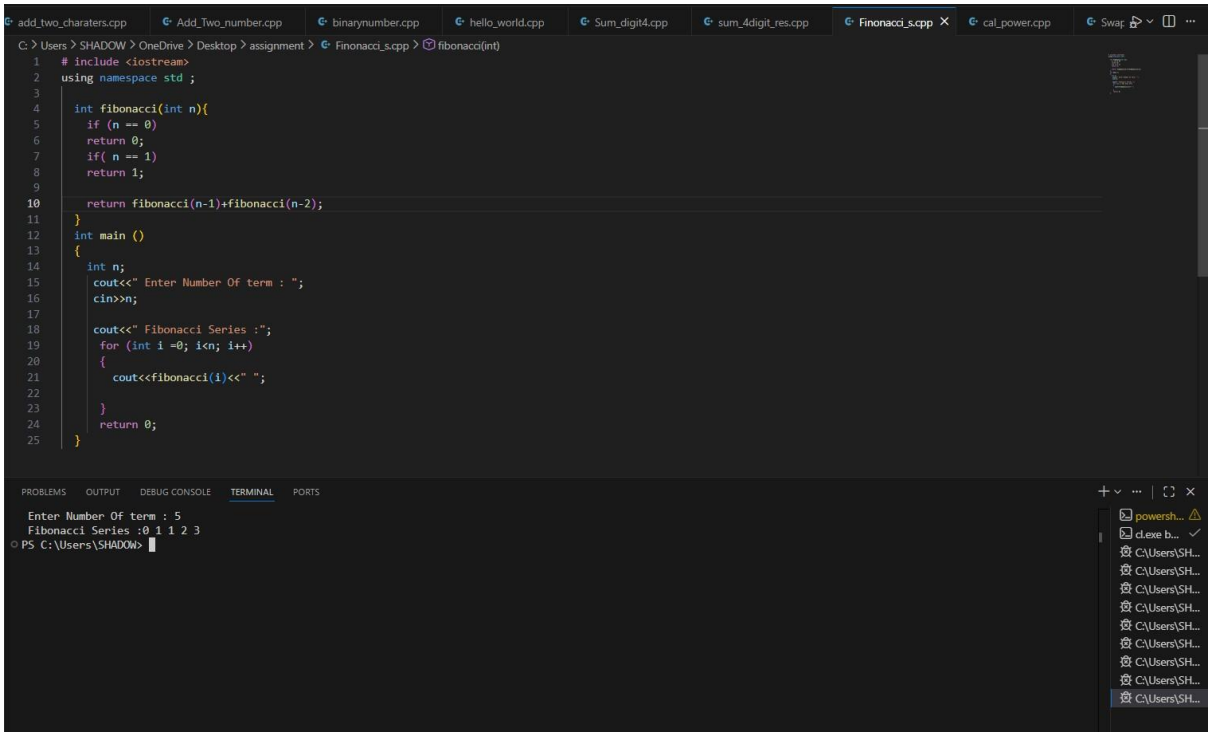
```
  return 0;
  return (n % 10) + sumdigit(n / 10);
 }

int main()
{
  int num ;
   cout<<" Enter $ digit Number : ";
   cin>>num;

   cout<<" Sum of digit = "<< sumdigit(num);

   return 0;
}
```



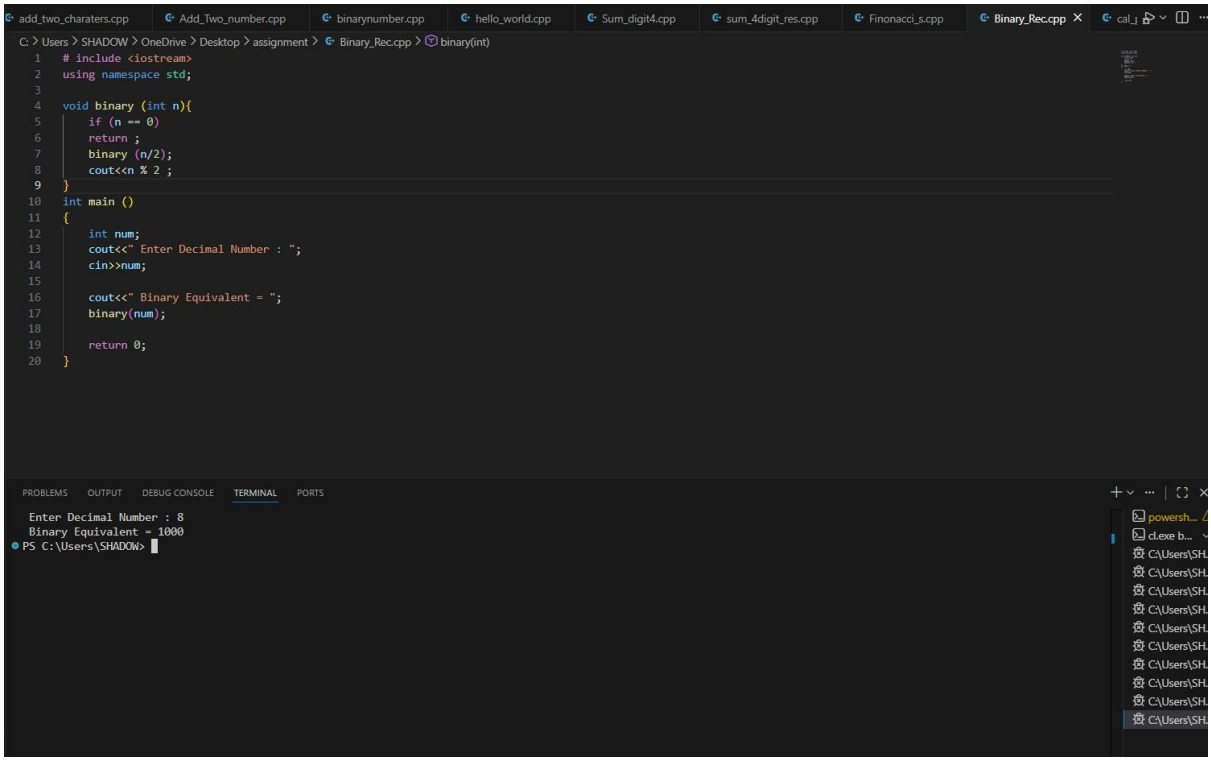
Q8] Fibonacci Series

```
# include <iostream>
using namespace std ;
```

```
int fibonacci(int n){
  if (n == 0)
  return 0;
  if( n == 1)
  return 1;

  return fibonacci(n-1)+fibonacci(n-2);
}
int main ()
{
  int n;
   cout<<" Enter Number Of term : ";
   cin>>n;

   cout<<" Fibonacci Series :";
    for (int i =0; i<n; i++)
    {
      cout<<fibonacci(i)<<" ";

    }
    return 0;
}
```



Q9] binary equivalent

```
# include <iostream>
using namespace std;

void binary (int n){
   if (n == 0)
   return ;
   binary (n/2);
   cout<<n % 2 ;
}
int main ()
{
   int num;
   cout<<" Enter Decimal Number : ";
   cin>>num;

   cout<<" Binary Equivalent = ";
```

```
    binary(num);


    return 0;

}
```